Generating Through the Noise: Progressive Denoising Diffusion for Infrared Small Target Detection

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***Abstract*—Infrared small target detection (IRSTD) aims to accurately identify dim and tiny targets in complex infrared scenes. However, weak target responses, low signal-to-noise ratios, and severe background interference make reliable target localization challenging, often resulting in missed detections and false alarms. Existing methods typically rely on deterministic feature learning to directly predict target masks, limiting the ability to progressively refine ambiguous target representations. To address these issues, we propose Progressive Denoising Diffusion (PDDiff), a conditional diffusion framework that reformulates IRSTD as a progressive generative denoising process. Specifically, a feature extractor is employed to capture discriminative infrared representations, followed by a conditional generator that constructs target guidance for subsequent denoising. Furthermore, Memory-guided Spectral Alignmen (MSA) are introduced to selectively refine conditional representations through frequency-domain memory selection, enabling adaptive guidance throughout the denoising process. By injecting the refined conditional information into the denoising network, PDDiff iteratively reconstructs accurate target masks from noisy representations. In addition, customized training and sampling strategies are developed to improve sparse target structure reconstruction and exploit intermediate predictions throughout the diffusion process, thereby mitigating structural errors and localization instability. Extensive experiments on two benchmark IRSTD datasets demonstrate the effectiveness of the proposed PDDiff in terms of detection accuracy and computational efficiency. The code will be published at https://github.com/jzw0707/PDDiff.**

***Index Terms*—Infrared small targte detection, conditional diffusion model, frequency selective modulation, diffusion training and sampling strategy**

I. Introduction

INFRARED small target detection (IRSTD) [1], [2] aims to accurately identify and localize dim and tiny targets in infrared imagery, serving as a fundamental technique in maritime surveillance, remote sensing, and environmental monitoring [3]–[6]. Unlike generic object detection, infrared small targets typically occupy only a few pixels (usually less than 16×16) and exhibit weak radiation responses, limited structural information and indistinct boundaries. These characteristics make them particularly susceptible to interference from complex backgrounds, such as clouds, ocean waves, and urban structures. Moreover, infrared imaging is inevitably affected by sensor noise, atmospheric disturbances, and imaging artifacts, which introduce additional clutter and noise into the observed images. Such interference may obscure weak target responses or produce target-like patterns, further increasing the difficulty of distinguishing genuine targets from background distractions. Consequently, accurately recovering subtle target information while suppressing complex background clutter and imaging noise remains a fundamental challenge for reliable IRSTD.

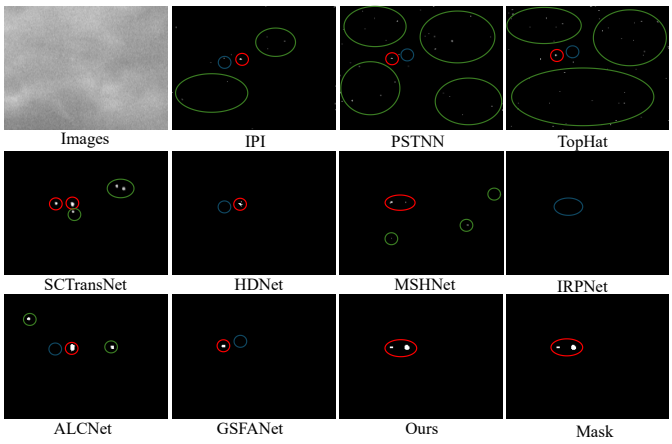


Fig. 1: Visual comparison of different IRSTD methods. Traditional methods suffer from severe background interference and false alarms (green circles), while deep learning methods still exhibit missed detections (blue circles). In contrast, the proposed PDDiff effectively suppresses false alarms and recovers genuine targets (red circles).

In general, traditional IRSTD methods primarily rely on handcrafted priors, such as local contrast enhancement [7], [8] and low-rank representation [9], [10], which are sensitive to complex background clutter and imaging noise. In contrast, deep learning methods have significantly advanced detection performance through hierarchical feature extraction and contextual representation learning. Specifically, CNN-based methods [11], [12] exploit multiscale feature interactions to enhance subtle target representations, while ViT-based approaches [13], [14] capture long-range dependencies through self-attention mechanisms. More recently, Mamba-based methods [15], [16] leverage state-space scaning for efficient global context aggregation, whereas SAM-based approaches [17], [18] explore pretrained segmentation priors for improved target discrimination.

Despite the significant progress achieved by deep learning-based methods, most existing approaches directly predict target masks from encoded features, where repeated downsampling and single-pass decoding may weaken subtle target responses. In principle, infrared small targets occupy only a few pixels and are easily overwhelmed by complex background clutter and sensor noise, making it difficult to distinguish genuine targets from target-like interference. Consequently, direct mask prediction may propagate ambiguous responses

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Manuscript received Month Day, Year; revised Month Day, Year.

into the final segmentation results, leading to missed detections and false alarms. This raises a fundamental question: **Can weak target representations be progressively refined rather than determined through a single prediction?** Diffusion models offer a promising alternative by reformulating target segmentation as a conditional generative denoising process. Through iterative refinement of noisy mask representations, diffusion models enable the progressive correction of uncertain predictions, potentially enhancing weak target responses while suppressing spurious background interference.

To enhance conditional guidance for progressive denoising, we introduce a Dynamic Multiscale Spatial Projector (DMSP) and a Memory-guided Spectral Alignment (MSA) module. Since weak infrared targets are easily obscured by background clutter and imaging noise, DMSP enhances target-sensitive spatial representations through dynamic multiscale projection, while MSA selectively retrieves relevant memory features based on frequency-domain similarity and performs phase alignment. Furthermore, MSA incorporates timestep information to adaptively modulate conditional features across denoising steps. Together, these modules provide complementary spatial-frequency guidance, facilitating accurate target mask generation while suppressing background interference.

Overall, our proposed Progressive Denoising Diffusion (PDDiff) establishes a progressive generative denoising paradigm for IRSTD. Specifically, PDDiff progressively reconstructs target masks from noisy representations under frequency-guided conditional information, enabling iterative refinement of weak target responses. As shown in Fig. 1, existing methods suffer from false alarms and missed detections under complex noise interference. In contrast, PDDiff progressively refines target predictions, effectively suppressing spurious responses and achieving more accurate segmentation results. The main contributions of this work are summarized as follows:

- We propose a conditional diffusion framework, termed PDDiff, that reformulates IRSTD as a progressive generative denoising process. Unlike conventional discriminative methods that directly predict target masks, PDDiff iteratively refines uncertain target representations, facilitating accurate target localization.
- We design a DMSP and a MSA to enhance conditional representations. DMSP adaptively aggregates multiscale spatial features, while MSA selectively retrieves relevant memory representations through frequency-domain similarity and performs phase alignment for progressive denoising.
- We develop customized training and sampling strategies, termed Sparse Mask Structure Perturbation (SMSP) and Timestep Prediction Aggregation (TPA), to improve diffusion-based IRSTD. Specifically, SMSP introduces structure-aware perturbations to enhance sparse target mask reconstruction during training, while TPA adaptively aggregates intermediate predictions across denoising steps to suppress false alarms during sampling.

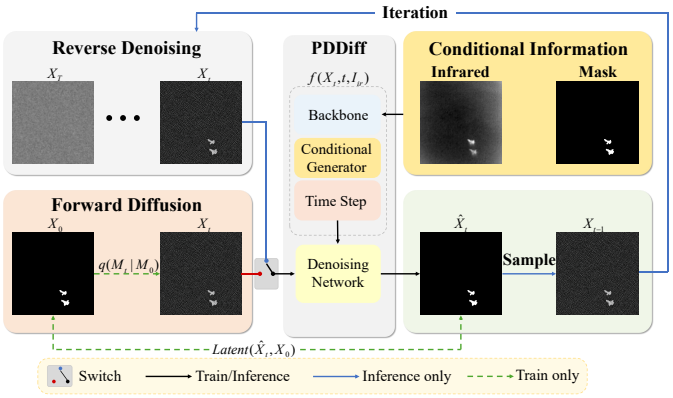


Fig. 2: Diffusion principle of our DiffIRSTD, which consists of forward diffusion and reverse denoising processes. During forward diffusion, random noise is progressively added to the mask X_0 to obtain a noisy mask X_t. Then, our DiffIRSTD, i.e., $f(X_t, t, I)$, is trained to predict the denoised mask $\hat{X}_t$ conditioned on the infrared image and diffusion timestep. During reverse denoising, DiffIRSTD iteratively refines the initial Gaussian noise X_T to generate the final masks.

II. Related Work

A. Infrared small target detection methods

Existing IRSTD methods can be broadly divided into traditional and deep learning-based approaches. Traditional methods, including Tophat [19], MPCN [20], PSTNN [21], and IPI [22], mainly rely on handcrafted priors, sparse representations, and low-rank tensor decomposition. However, their performance is often limited by complex background clutter and weak target responses. Deep learning methods have achieved significant progress through various architectures. CNN-based methods, such as ACM [23] and DNANet [11], employ contextual modeling and attention mechanisms to enhance target representations. Transformer-based methods, such as RKformer [24] and BMPTransNet [25], explore hybrid architectures for long-range dependency modeling. Frequency-aware methods, including Dim2Clear [26] and GSFANet [27], exploit spatial-frequency interactions to improve feature discrimination. More recently, mamba-based methods, such as MiMISTD [28] and SAMamba [29], leverage Mamba architectures for efficient feature modeling, while FAMENet [30] investigate expert-based modeling and multimodal guidance, respectively.

However, these IRSTD methods adopt a single-step prediction paradigm, directly mapping ambiguous target responses to segmentation masks without explicit error correction, often leading to missed detections and false alarms. In contrast, diffusion models offer a progressive denoising mechanism that enables iterative refinement of uncertain predictions under complex background clutter. Nevertheless, directly applying conventional diffusion models to IRSTD remains challenging. During training, conventional Gaussian perturbations overlook the sparse structure of target masks, while during sampling, relying solely on the final prediction fails to exploit useful information from intermediate denoising steps. To address these limitations, we introduce SMSP and TPA for training

and sampling, respectively. SMSP enhances structure-aware denoising of sparse targets, while TPA aggregates intermediate predictions to improve detection accuracy.

B. Diffusion Models

Diffusion models have attracted increasing attention due to their remarkable generative capabilities and ability to model complex data distributions through iterative denoising. Early studies, such as DDPM [31] and latent diffusion models [32], demonstrated their effectiveness in high-quality image generation. Subsequently, diffusion models have been extended to low-level vision tasks, including image super-resolution and restoration. Representative methods, such as SR3 [33] and DiffIR [34], progressively reconstruct images from noisy representations through conditional denoising. More recently, diffusion models have shown considerable potential in high-level vision tasks. For example, DiffusionDet [35] formulates object detection as a denoising process over bounding boxes, while DiMSOD [36] employs diffusion to generate accurate saliency masks for multimodal salient object detection. Despite the growing success of diffusion models in various vision tasks, recent studies have explored their application to IRSTD. Du et al. [37] proposed ISTD-diff, which formulates IRSTD as a conditional mask generation process through iterative denoising. Li et al. [38] introduced posterior distribution modeling with wavelet-domain isolation to suppress infrared noise interference. Zhang et al. [39] further proposed DEDM-Net, which employs dual-path diffusion modeling for mask generation.

Given the progressive denoising nature of diffusion models, we argue that preserving sparse target structures during training and exploiting intermediate predictions during sampling are essential for IRSTD. However, conventional diffusion models [40], [41] primarily rely on Gaussian perturbations, which fail to explicitly account for the sparse structures of target masks, limiting their ability to correct structural errors during mask generation. Moreover, relying solely on the final denoising output overlooks valuable information from intermediate predictions, potentially leading to inaccurate segmentation. To address these limitations, we introduce SMSP to simulate structural deviations in sparse target masks and enhance structure reconstruction during training, while TPA adaptively aggregates intermediate predictions to improve the robustness of generated masks during sampling.

III. Methodology

A. Diffusion Principle

Diffusion models formulate data generation as a probabilistic process that progressively transforms random noise into structured representations through a sequence of denoising steps. As illustrated in Fig. 2, the diffusion process consists of two stages: forward diffusion and reverse denoising. In the forward diffusion process, the ground-truth mask x_0 is progressively corrupted by Gaussian noise over T time steps, producing a sequence of noisy masks $x_1, \ldots, x_T$. This Markov process [42] is formulated as:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbf{I}\right), \tag{1}$$

where $\alpha_t = 1 - \beta_t$, $\beta_t \in (0,1)$ denotes the variance schedule, and $\mathbf{I}$ is the identity matrix. Based on the Markov property, the noisy mask x_t can be directly sampled from x_0 as:

$$q(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I}\right), \tag{2}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. In the reverse denoising process, a neural network f_θ progressively reconstructs the target mask from Gaussian noise x_T by estimating the clean mask at each time step. The reverse transition is defined as:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \tilde{\beta}_t\mathbf{I}\right), \tag{3}$$

where the variance $\tilde{\beta}_t$ is given by:

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}(1-\alpha_t), \tag{4}$$

and the mean $\mu_\theta(x_t, t)$ is calculated as:

$$\begin{aligned}\mu_\theta(x_t, t) = &\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t \\ &+ \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\hat{x}_0^{(t)},\end{aligned} \tag{5}$$

where $\hat{x}_0^{(t)}$ denotes the clean mask estimated by the denoising network at time step t.

Specifically, in our PDDiff, the reverse process is conditioned on the input infrared image I_{IR}. The conditional representations $\mathbf{C}$ are extracted and refined by the feature extractor, DMSP, and MSA, respectively. The conditional denoising process is formulated as:

$$p_\theta(x_{t-1}|x_t, \mathbf{C}) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t, \mathbf{C}), \tilde{\beta}_t\mathbf{I}_{\mathrm{ir}}\right), \tag{6}$$

where $\mathbf{C}$ represents the conditional information derived from I_{ir}, and the predicted clean mask is obtained as $\hat{x}_0^{(t)} = f_\theta(x_t, t, \mathbf{C})$. Guided by these conditional representations, PDDiff progressively refines noisy masks to recover accurate target predictions.

B. Overall network

As illustrated in Fig. 2, our PDDiff consists of four key components: a backbone, a conditional generator, a diffusion timestep modulation stage, and a denoising network. Given an infrared image $\mathbf{I}_{\mathrm{ir}}$, we adopt the SAM2 [43] encoder as the backbone to extract hierarchical features. Specifically, the pretrained Hiera blocks are frozen, while the trainable adapters, each consisting of two linear layers, are inserted into each stage for domain adaptation. The extracted four-level features are denoted as $\mathbf{S}_i (i \in 0, 1, 2, 3)$.

In the conditional generator, we introduce a Dynamic Multiscale Spatial Projector (DMSP) to refine the extracted features through adaptive multiscale spatial projection, generating conditional representations $\mathbf{S}_i'$. To adapt conditional features to varying noise levels across diffusion steps, we introduce timestep into the Multi-head Cross Attention (MCA), where the timestep embedding t serves as the query and $\mathbf{S}_i'$ as the keys and values, generating timestep representations $\mathbf{S}_i^t$. Meanwhile, these features are progressively aggregated through Fusion blocks and concatenated into a unified representation,

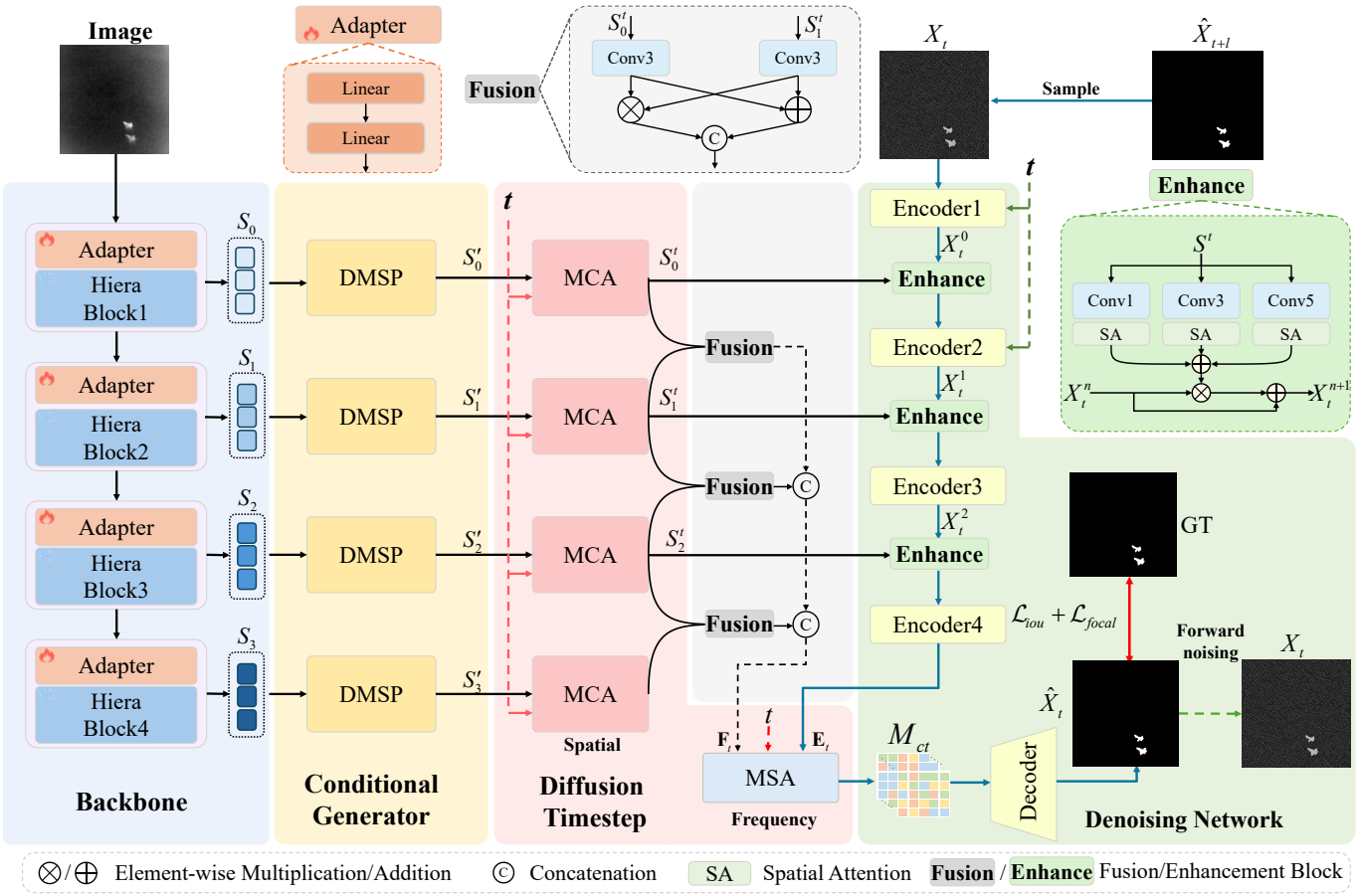


Fig. 3: The architecture of our PDDiff. The feature extractor captures hierarchical infrared features, which are refined by DMSP and MSA to generate effective conditional information. The denoising network progressively reconstructs target masks under conditional guidance. Additionally, SMSP enhances sparse target learning during training, while TPA aggregates intermediate predictions during sampling to improve detection accuracy.

which is further refined by MSA through frequency-domain memory selection and phase alignment, generating refined conditional features $\mathbf{M}_{ct}$.

In the denoising network, the noisy mask X_t is processed through a hierarchical encoder, where the corresponding $\mathbf{S}_i^t$ are injected via Enhancement blocks. Meanwhile, $\mathbf{M}_{ct}$ is incorporated into the decoder to provide complementary frequency-guided information. The decoder then predicts the target mask $\hat{X}_t$, which guides subsequent sampling steps for progressive target reconstruction.

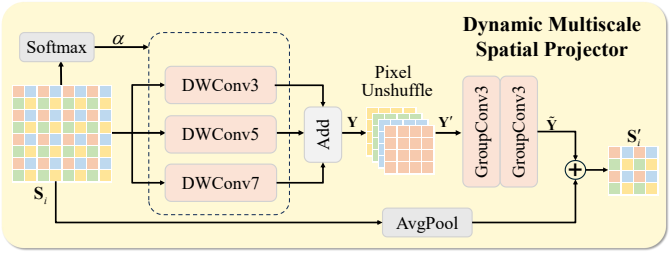


Fig. 4: The architecture of the proposed Dynamic Multiscale Spatial Projector (DMSP).

C. Conditional Generator and Diffusion Timestep

1) Dynamic Multiscale Spatial Projector (DMSP): To improve the quality of conditional information, we introduce a DMSP into the conditional generator, which adaptively integrates multiscale spatial features to preserve edge structures, thereby providing discriminative guidance for progressive denoising. As shown in Fig. 4, given an input feature $\mathbf{S}_i \in \mathbb{R}^{C\times H\times W}$, DMSP first applies three depthwise convolution with kernel sizes k of 3, 5, and 7 to capture spatial responses at different scales. A channel-wise attention mechanism then predicts adaptive fusion weights α_k through softmax activation, yielding:

$$\mathbf{Y} = \sum_{k\in\{3,5,7\}} \alpha_k \odot \mathrm{DWConv}_k(\mathbf{S}_i), \tag{7}$$

where $\odot$ denotes element-wise multiplication. The adaptive filtering balances fine-grained target preservation across different spatial scales. Subsequently, a pixel-unshuffle operation reorganizes local spatial information $\mathbf{Y}'$ into the channel dimension, which preserves local sub-pixel information during

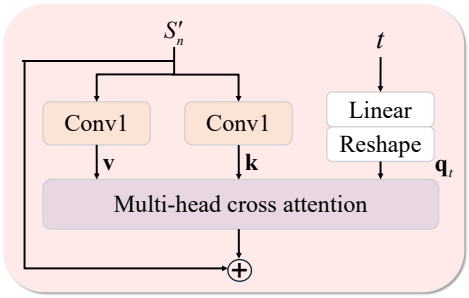


Fig. 5: The architecture of the proposed Multi-head Cross Attention (MCA).

spatial resolution reduction. This process is described as follows:

$$\mathbf{Y}' = \mathrm{PixelUnshuffle}(\mathbf{Y}) \in \mathbb{R}^{4C\times\frac{H}{2}\times\frac{W}{2}}, \tag{8}$$

The resulting features are further processed by two grouped convolutions to restore the channel dimension and refine local representations $\widetilde{\mathbf{Y}}$. Meanwhile, a residual branch employs average pooling to retain low-frequency structural information. The final projected feature $\mathbf{S}'_i$ is obtained as:

$$\mathbf{S}'_i = \mathrm{ReLU}\left(\widetilde{\mathbf{Y}} + \mathrm{AvgPool}(\mathbf{S}_i)\right). \tag{9}$$

Through adaptive spatial filtering, DMSP suppresses irrelevant background responses while preserving subtle target structures, providing reliable conditional representations for subsequent timestep modulation.

2) Multi-head Cross Attention (MCA): To adapt conditional information to varying noise levels across diffusion steps, we introduce a MCA module for timestep feature modulation. Given the conditional feature $\mathbf{S}'_n \in \mathbb{R}^{C\times H\times W}$ and diffusion timestep t, MCA first projects $\mathbf{S}'_n$ into keys $\mathbf{k}$ and values $\mathbf{v}$ through two 1×1 convolutions. Meanwhile, the timestep embedding is transformed through a linear layer and reshaped into the query $\mathbf{q}_t$. The resulting representations are fed into multi-head cross-attention to adaptively modulate the conditional features according to the current diffusion step. Since the timestep embedding contains no explicit spatial information, MCA performs timestep-dependent channel modulation without introducing unnecessary spatial attention. A residual connection is then employed to preserve the original conditional information, yielding:

$$\mathbf{S}^t_n = \mathbf{S}'_n + (\mathrm{MCA}(\mathbf{q}_t, \mathbf{k}, \mathbf{v})), \tag{10}$$

where $\mathrm{MCA}(\cdot)$ denotes multi-head cross-attention. The resulting timestep-dependent features $\mathbf{S}^t_n$ are subsequently aggregated through multiscale Fusion blocks and concatenation to provide comprehensive conditional information for the following MSA module.

3) Memory-guided Spectral Alignment (MSA): Direct aggregation of multiscale features may weaken target discrimination across feature levels [44]. To address this issue, we introduce an MSA module, which selectively retrieves relevant memory representations through frequency similarity and performs phase alignment to enhance conditional guidance. As illustrated in Fig. 6, MSA takes the fused conditional feature

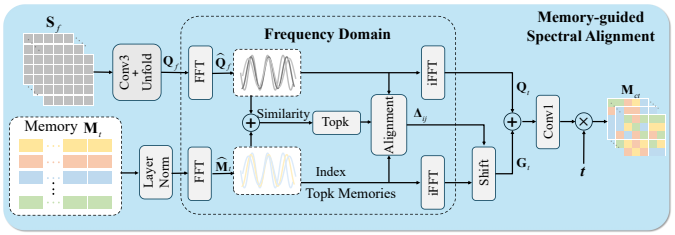


Fig. 6: The architecture of the proposed Memory-guided Spectral Alignment (MSA).

$\mathbf{S}_f$ and the deep feature $\mathbf{E}_t$ from the final encoder stage as inputs. Specifically, $\mathbf{S}_f$ is projected through a 3×3 convolution and unfolded into query tokens $\mathbf{Q}_f \in \mathbb{R}^{N\times D}$, while $\mathbf{E}_t$ provides memory tokens $\mathbf{M}_t \in \mathbb{R}^{L\times D}$, where N and L denote the numbers of query and memory tokens, respectively. The memory bank contains $[256, 128, 64, 32]$ patches across four levels, with the corresponding Top$K$ retrieval numbers set to $[8, 8, 8, 4]$. Both representations are then transformed into the frequency domain:

$$\hat{\mathbf{Q}}_f = \mathcal{F}(\mathbf{Q}_f), \quad \hat{\mathbf{M}}_t = \mathcal{F}\left(\mathrm{LN}(\mathbf{M}_t)\right), \tag{11}$$

where $\mathcal{F}(\cdot)$ denotes the fast Fourier transform (FFT), and $\mathrm{LN}(\cdot)$ represents layer normalization. To retrieve relevant memory patterns, we calculate the cosine similarity between their frequency magnitude representations:

$$\mathrm{sim}_{ij} = \frac{\langle|\hat{\mathbf{q}}_i|, |\hat{\mathbf{m}}_j|\rangle}{\||\hat{\mathbf{q}}_i|\|_2 \, \||\hat{\mathbf{m}}_j|\|_2}, \tag{12}$$

where $\hat{\mathbf{q}}_i$ and $\hat{\mathbf{m}}_j$ denote the frequency representations of the i-th query and j-th memory token of $\hat{\mathbf{Q}}_f$ and $\hat{\mathbf{M}}_t$, respectively. Based on the similarity matrix, the top$K$ memory tokens are selected to provide relevant structural information while suppressing unrelated background responses. Since the retrieved memory features may exhibit spatial offsets relative to the current query, we further perform phase correlation to estimate their relative displacement:

$$\mathbf{\Delta}_{ij} = \arg\max_{\boldsymbol{\delta}} \mathcal{F}^{-1}\left(\frac{\hat{\mathbf{q}}_i \odot \hat{\mathbf{m}}^*_j}{|\hat{\mathbf{q}}_i \odot \hat{\mathbf{m}}^*_j| + \epsilon}\right), \tag{13}$$

where $(\cdot)^*$ denotes complex conjugation, ϵ ensures numerical stability, and $\mathbf{\Delta}_{ij}$ represents the estimated spatial displacement. The selected memory features are then transformed back into the spatial domain and shifted accordingly. The aligned representations are aggregated as:

$$\mathbf{g}_i = \sum_{j\in\mathcal{K}_i} w_{ij}\,\mathrm{Shift}\left(\mathcal{F}^{-1}(\hat{\mathbf{m}}_j), \mathbf{\Delta}_{ij}\right), \tag{14}$$

where $\mathcal{K}_i$ denotes the selected top$K$ memory indices and w_{ij} represents the normalized similarity weight. Finally, the query features reconstructed via iFFT are combined with the aligned memory features through element-wise addition, followed by a 1×1 convolution and timestep modulation to generate the refined conditional representation $\mathbf{M}_{ct}$. The fused features are further projected and modulated by the timestep embedding:

$$\mathbf{M}_{ct} = \mathrm{Conv}_{1\times 1}\left(\mathbf{Q}_t + \mathbf{G}_t\right) \odot \mathbf{t}, \tag{15}$$

where $\mathbf{t}$ denotes the timestep embedding and $\odot$ represents element-wise multiplication. Through frequency-guided memory selection and phase alignment, MSA integrates multiscale conditional features with deep encoder representations and incorporates timestep information to provide adaptive guidance across diffusion steps.

D. Denoising Network

As illustrated in Fig. 3, the denoising network adopts a four-level encoder-decoder architecture to progressively reconstruct target masks from noisy representations. The encoder consists of four convolutional stages, where timestep embeddings are incorporated to adapt feature extraction across different noise levels. To provide target-sensitive guidance, Enhancement blocks are inserted after the first three encoder stages to refine intermediate features using conditional representations $\mathbf{S}_i^t$ modulated by the diffusion timestep. Specifically, each Enhancement block employs three parallel convolutional branches with kernel sizes of 1, 3, and 5, followed by spatial attention (SA) to capture complementary spatial responses. The resulting attention maps are aggregated to modulate the encoder features through residual enhancement. The deepest encoder features are further utilized by MSA to generate frequency-guided conditional information $\mathbf{M}_{ct}$. The decoder then integrates $\mathbf{M}_{ct}$ to reconstruct the predicted target mask $\hat{\mathbf{x}}_t$, which is used for subsequent denoising steps.

As for loss function, we employ a hybrid loss function combining Focal loss and IoU loss. The overall loss function is formulated as:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{focal} + \lambda_2 \mathcal{L}_{iou}, \tag{16}$$

where λ_1 and λ_2 are the weighting coefficients for Focal loss and IoU loss, empirically set to 2 and 5, respectively. The combined objective enhances the discrimination of sparse targets while improving the spatial consistency of predicted masks.

E. Training Strategy

Sparse Mask Structure Perturbation (SMSP). Conventional diffusion models typically corrupt ground-truth masks through Gaussian noise injection, which primarily introduces pixel-level disturbances without explicitly modeling structural errors in target predictions. Consequently, the denoising network may rely excessively on noisy mask structures, limiting its ability to correct inaccurate predictions during sampling, particularly for sparse infrared targets. To address this issue, we introduce SMSP to improve the robustness of target mask reconstruction. Specifically, given a ground-truth mask $\mathbf{x}_0$, SMSP randomly perturbs its target structures through local contour deformation, erosion, dilation, and spatial shifting, generating a structurally corrupted mask $\widetilde{\mathbf{x}}_0$. This process is formulated as:

$$\widetilde{\mathbf{x}}_0 = \mathcal{P}_\xi(\mathbf{x}_0), \tag{17}$$

where $\mathcal{P}_\xi(\cdot)$ denotes the stochastic structure perturbation operation controlled by random parameters ξ. Subsequently, a diffusion timestep t is uniformly sampled from $\{1, \ldots, T\}$, and

Algorithm 1 Training of PDDiff

Require: Dataset $\mathcal{D}$, diffusion steps T, parameters θ
Ensure: Optimized parameters θ
repeat
  $(\mathbf{I}_{\mathrm{IR}}, \mathbf{x}_0) \sim \mathcal{D}$
  $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
  $\widetilde{\mathbf{x}}_0 \leftarrow \mathrm{SMSP}(\mathbf{x}_0)$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\mathbf{x}_t \leftarrow \sqrt{\bar{\alpha}_t}\widetilde{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$
  $\hat{\mathbf{x}}_0^{(t)} \leftarrow f_\theta(\mathbf{x}_t, t, \mathbf{I}_{\mathrm{ir}})$
  $\mathcal{L} \leftarrow 2\mathcal{L}_{focal}(\hat{\mathbf{x}}_0^{(t)}, \mathbf{x}_0) + 5\mathcal{L}_{iou}(\hat{\mathbf{x}}_0^{(t)}, \mathbf{x}_0)$
  Update θ using Adam with $\nabla_\theta \mathcal{L}$
until maximum training epochs are reached

Algorithm 2 Sampling of PDDiff

Require: Infrared image $\mathbf{I}_{\mathrm{ir}}$, sampling steps T, trained parameters θ
Ensure: Aggregated confidence map $\mathbf{P}$
$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
Initialize $\mathcal{P} \leftarrow [\,]$
for $t = T, T-1, \ldots, 1$ **do**
  $\hat{\mathbf{x}}_0^{(t)} \leftarrow f_\theta(\mathbf{x}_t, t, \mathbf{I}_{\mathrm{ir}})$
  Append $\hat{\mathbf{x}}_0^{(t)}$ to $\mathcal{P}$
  if $t > 1$ **then**
    $\boldsymbol{\mu}_t \leftarrow \mu_\theta(\mathbf{x}_t, \hat{\mathbf{x}}_0^{(t)}, t)$
    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
    $\mathbf{x}_{t-1} \leftarrow \boldsymbol{\mu}_t + \sqrt{\tilde{\beta}_t}\boldsymbol{\epsilon}$
  end if
end for
$\mathbf{P} \leftarrow \mathrm{TPA}(\mathcal{P})$
return $\mathbf{P}$

Gaussian noise is injected into the perturbed mask according to the predefined noise schedule:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\widetilde{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{18}$$

where $\bar{\alpha}_t$ denotes the cumulative noise schedule. As illustrated in Alg. 1, conditioned on the infrared image $\mathbf{I}_{\mathrm{ir}}$, the denoising network f_θ predicts the clean target mask as $\hat{\mathbf{x}}_0^{(t)} = f_\theta(\mathbf{x}_t, t, \mathbf{I}_{\mathrm{ir}})$. The original ground-truth mask $\mathbf{x}_0$ is used for supervision, encouraging the network to reconstruct accurate target structures from noisy and spatially perturbed inputs rather than reproducing corrupted representations. In addition, random cropping, flipping, and intensity jittering are employed to enhance training diversity. By integrating structural perturbations into the Gaussian diffusion process, SMSP improves the robustness of mask reconstruction and facilitates the recovery of sparse target structures during progressive denoising.

F. Sampling Strategy

Timestep Prediction Aggregation (TPA). During sampling, PDDiff progressively reconstructs target masks from Gaussian noise through iterative denoising. Although the final prediction generally provides refined target structures,

TABLE I: Performance comparison of model-driven and data-driven methods on SIRST-UAVB and SIRST4 in terms of IoU (%), F_1 (%), P_d (%), and F_a (10^{-6}), along with the number of parameters (M), computational complexity (GFLOPs), and inference speed (FPS). Red indicates the best performance, blue indicates the second-best.

Method	SIRST-UAVB				SIRST4				Params↓	GFLOPs↓	FPS↑
	IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓			
Model-driven Methods											
TopHat [19]	10.28	73.11	58.40	1281	5.53	62.73	51.65	1873	-	-	-
WSLCM [45]	7.65	66.49	53.91	2763	6.39	81.36	77.05	1874	-	-	-
TLLCM [46]	6.38	59.76	50.92	4478	4.75	76.38	69.51	3281	-	-	-
IPI [22]	27.84	80.45	76.80	27.49	17.74	73.86	64.47	44.68	-	-	-
NRAM [47]	9.55	72.63	65.19	127.84	12.30	67.49	62.09	93.45	-	-	-
RIPT [47]	13.85	74.55	62.47	110.78	14.79	68.76	61.39	99.81	-	-	-
PSTNN [21]	22.71	79.32	74.56	83.38	18.79	73.26	69.11	106.40	-	-	-
Data-driven Methods											
ACM [23]	58.16	80.33	71.29	8.69	69.77	92.13	79.60	21.32	0.39	0.40	109.8
AGPCNet [48]	56.89	77.02	63.83	11.94	68.48	88.72	75.15	28.62	12.36	43.18	19.30
ALCNet [49]	38.31	78.08	55.40	10.63	58.83	92.26	74.08	28.68	0.43	0.38	202.69
DNANet [11]	64.85	82.12	78.68	1.87	77.24	95.41	87.16	6.04	4.70	14.26	9.36
GSFANet [27]	62.30	89.38	76.77	3.61	71.87	96.02	83.63	12.82	2.99	5.25	9.56
HDNet [50]	64.89	83.64	78.71	2.36	77.43	95.19	87.28	9.69	3.68	11.20	16.70
IRPNet [51]	70.47	96.48	83.11	3.84	77.56	96.81	87.53	17.39	32.34	26.63	50.14
ISNet [52]	56.19	86.17	71.95	6.21	68.57	93.76	81.35	28.67	0.97	30.62	37.02
ISTDU-Net [53]	61.72	88.70	73.63	8.82	70.51	92.18	79.64	22.43	2.75	7.94	37.00
MSHNet [54]	60.96	80.44	75.74	3.54	74.57	94.14	85.44	8.47	4.07	19.27	12.63
RDANet [44]	68.19	96.37	81.89	5.34	75.34	94.03	85.28	11.49	4.17	25.24	15.00
RDIAN [55]	59.38	87.44	71.98	7.22	69.08	93.61	80.94	22.57	0.22	3.72	56.00
SCTransNet [13]	63.56	88.20	77.72	4.25	77.13	96.69	87.09	18.70	11.33	20.24	14.95
SPKAN [56]	70.39	96.31	82.76	4.48	75.80	96.17	86.24	19.85	9.52	8.18	70.15
IRSTD-Diff [38]	62.47	88.07	77.32	6.71	74.68	93.36	84.49	18.26	53.48	104.64	22.17
UIUNet [12]	62.37	79.43	78.74	6.15	65.98	96.84	79.50	42.88	50.54	54.43	31.45
Ours	71.96	96.80	83.70	2.53	79.45	97.14	88.55	6.34	43.71	123.98	25.74

intermediate predictions also contain valuable information for identifying infrared small targets. However, these predictions may exhibit different levels of uncertainty across sampling steps [57]. Inspired by prediction aggregation in segmentation [58], we employ TPA to exploit intermediate predictions. Specifically, given the predicted probability maps $\{\mathbf{P}_j\}_{j=1}^{K}$ from K sampling steps, TPA first generates binary masks $\mathbf{B}_j$ through adaptive thresholding. These masks are then combined through weighted voting to identify reliable target regions. To account for differences in prediction stability, the aggregation weights are determined by the discrepancy between consecutive predictions:

$$w_j = \frac{1}{\|\mathbf{P}_j - \mathbf{P}_{j-1}\|_2^2 + \epsilon}, \tag{19}$$

where $j = 2, \ldots, K$, ϵ ensures numerical stability, and $w_1 = w_2$ for $K \geq 2$. Smaller discrepancies indicate more consistent predictions, which are assigned larger weights. Finally, the weighted binary masks determine the candidate target regions, while the corresponding probability maps provide refined confidence scores:

$$\mathbf{P} = \mathrm{Round}\left(\frac{\sum_{j=1}^{K} w_j \mathbf{B}_j}{\sum_{j=1}^{K} w_j}\right) \odot \frac{\sum_{j=1}^{K} w_j \mathbf{P}_j}{\sum_{j=1}^{K} w_j}, \tag{20}$$

where $\odot$ denotes element-wise multiplication. As illustrated in Alg. 2, the denoising network f_θ iteratively predicts clean target masks from noisy representations conditioned on the infrared image $\mathbf{I}_{\mathrm{ir}}$. The predictions generated at different sampling steps are collected and aggregated by TPA to obtain the final confidence map $\mathbf{P}$. By exploiting prediction consistency across denoising steps, TPA suppresses unstable target responses and improves the reliability of mask generation without introducing additional denoising iterations.

IV. Experiments and Results

A. Experimental Setup

Experiments are conducted on two public benchmarks, SIRST-UAVB [59] and SIRST4 [60], containing 3,000 and 3,352 infrared images, respectively. SIRST4 integrates several widely used datasets, including SIRSTv2, NUDT-SIRST, and IRSTD-1K, covering diverse backgrounds and target scales. SIRST-UAVB mainly contains UAV-based urban scenes characterized by complex clutter. The dataset split follows the default setting. For evaluation, we adopt standard metrics widely used in IRSTD, including intersection over union (IoU), F1 measure, probability of detection (P_d), and false alarm rate (F_a).

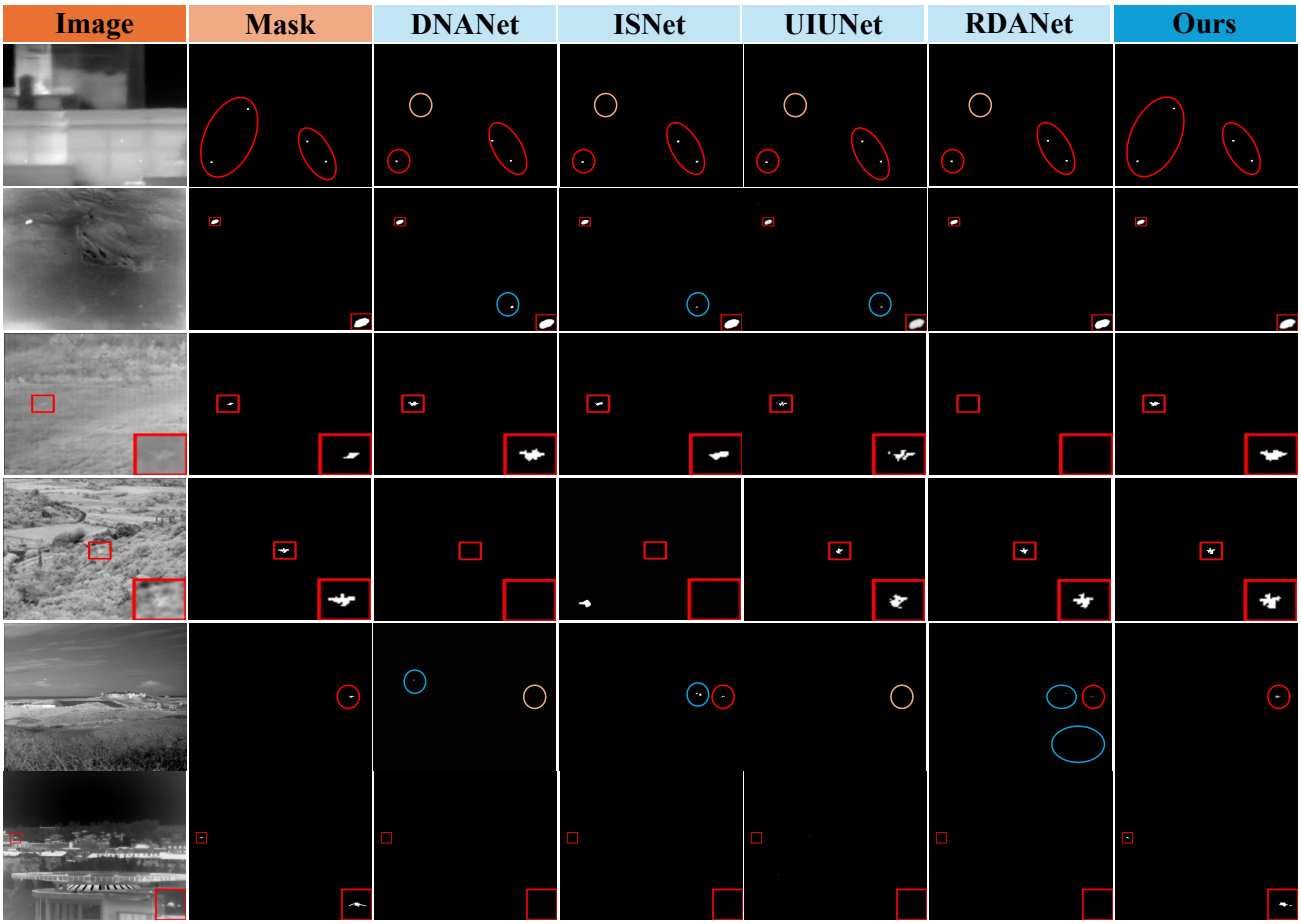


Fig. 7: Qualitative results of different methods on the SIRST4 and SIRST-UAVB dataset. Red circles denote correct detections, blue circles denote false alarms, and orange circles denote miss detections.

We implement our PDDiff using PyTorch on two NVIDIA 5880 GPUs. The pretrained SAM2-Tiny encoder with trainable adapters is adopted as the backbone, while the encoder parameters remain frozen. All infrared images and corresponding masks are resized to 512×512. We employ the Adam optimizer with an initial learning rate of 1×10^{-4}, which is gradually reduced to 1×10^{-7} using a CosineAnnealingLR scheduler. The model is trained for 300 epochs with a batch size of 8. During training, SMSP introduces random contour perturbations into target masks before Gaussian noise injection to enhance structure denoising. Random cropping, intensity jittering, and flipping are further applied for data augmentation. The network is optimized using a combination of Focal loss and IoU loss, with weighting coefficients $\lambda_1 = 2$ and $\lambda_2 = 5$, respectively. During sampling, TPA aggregates intermediate predictions across denoising steps to improve detection robustness.

B. Compared with SOTA methods

1) Quantitative Results: As reported in Table I, PDDiff achieves the best IoU, P_d, and F1 on SIRST-UAVB, reaching 71.96%, 96.80%, and 83.70%, respectively. Compared with the second-best IRPNet, PDDiff improves IoU, P_d, and F1 by 1.49%, 0.32%, and 0.59%, respectively. Notably, PDDiff also surpasses the diffusion-based method IRSTD-Diff by 9.49% IoU and 8.73% P_d, demonstrating its effectiveness in accurately identifying infrared small targets. Although DNANet achieves the lowest F_a of 1.87, PDDiff maintains a relatively low F_a of 2.53 while delivering superior detection accuracy.

On SIRST4, PDDiff consistently achieves the best IoU, P_d, and F1 of 79.45%, 97.14%, and 88.55%, respectively. Compared with IRPNet, the second-best method in IoU, PDDiff improves IoU by 1.89%. It also exceeds UIUNet by 0.30% in P_d and HDNet by 1.27% in F1. Furthermore, compared with IRSTD-Diff, PDDiff improves IoU and P_d by 4.77% and 3.78%, respectively. Overall, the results on the two datasets further demonstrate the effectiveness of diffusion progressive denoising for IRSTD, which is well suited to the noise-sensitive characteristics of infrared imaging, enabling gradual suppression of background clutter and imaging noise while improving the discrimination of weak targets from target-like interference.

2) Tradeoff Analysis: To further evaluate the computational efficiency of PDDiff, we analyze the tradeoff between detection accuracy and inference speed. As reported in Table I, PDDiff achieves the highest detection accuracy with 43.71M parameters and an inference speed of 25.74 FPS. Compared with the IRSTD-Diff, PDDiff achieves higher accuracy with

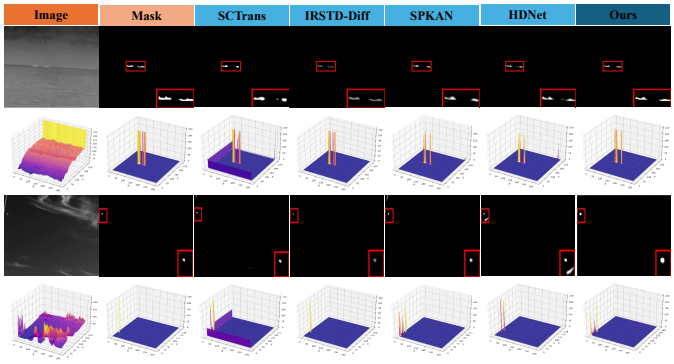


Fig. 8: Qualitative results of different methods on the SIRST4 dataset.

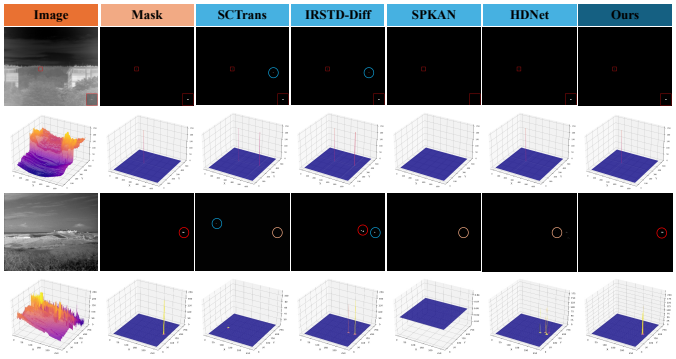


Fig. 9: Qualitative results of different methods on the SIRST-UAVB dataset.

fewer parameters and faster inference. Although iterative denoising introduces additional computational overhead, the number of sampling steps can be flexibly adjusted to balance detection performance and inference efficiency. This flexibility makes PDDiff a competitive candidate for real-time IRSTD applications while retaining the advantages of progressive target refinement.

3) Visual Results: Figs. 7-9 present qualitative comparisons of different IRSTD methods under complex infrared scenes. In Fig. 7, weak and densely distributed targets are easily confused with background interference, where DNANet, ISNet, and UIUNet exhibit missed detections and false alarms. In contrast, PDDiff better preserves subtle target structures while suppressing spurious background responses. Fig. 8 further compares detection results in maritime and complex sky backgrounds. Although methods such as SCTransNet and IRSTD-Diff can identify certain targets, they still suffer from incomplete target responses or background interference. PDDiff produces more complete target masks with fewer irrelevant activations. Meanwhil, the corresponding 3D response maps also show that PDDiff generates more concentrated target peaks with suppressed background responses.

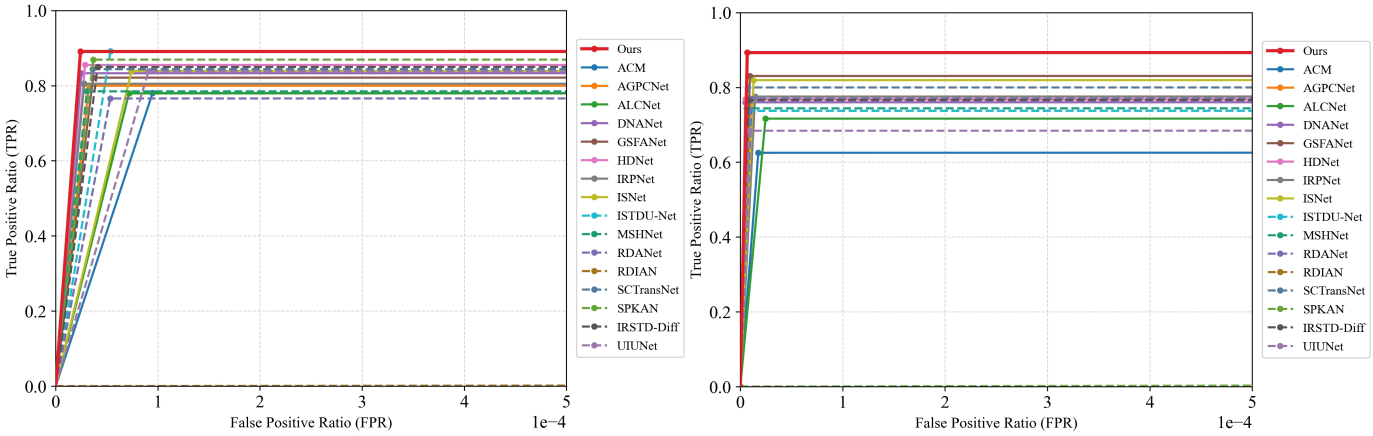
4) ROC Curve: Fig. 10 presents the ROC curves of different methods on SIRST4 and SIRST-UAVB. PDDiff consistently achieves superior detection performance on both datasets, maintaining higher true positive rates (TPR) under low false positive ratios (FPR). The improvements are particularly evident on SIRST-UAVB, demonstrating the effectiveness of progressive denoising in distinguishing weak infrared targets from background interference.

C. Ablation Study

1) Integrated Framework.: Table II reports the ablation results of the proposed components on SIRST-UAVB and SIRST4. Compared with the baseline, introducing the denoising network improves IoU by 4.84% and 3.12%, and P_d by 5.22% and 2.09% on SIRST-UAVB and SIRST4, respectively. These improvements demonstrate that diffusion-based progressive denoising facilitates iterative refinement of weak target predictions compared with direct mask prediction. Incorporating DMSP further increases IoU to 69.74% and 77.19%, indicating that dynamic multiscale spatial projection enhances generation conditional representations. With MCA, IoU improves to 71.23% and 78.81%, while P_d reaches 96.39% and 97.06%, respectively, demonstrating the effectiveness of timestep conditional modulation. Finally, MSA achieves the best overall performance, increasing IoU to 71.96% and 79.45% while reducing F_a from 4.48 to 2.53 and from 8.27 to 6.34, respectively. By selectively retrieving relevant memory features through Top-K frequency similarity and performing alignment, MSA enhances target discrimination while suppressing irrelevant background responses.

The heatmap results in Fig. 11 further demonstrate the effectiveness of the proposed components. The baseline exhibits widespread background activations, indicating severe interference from imaging scenes. By introducing diffusion denoising and DMSP, background responses are gradually suppressed while weak target features become more distinguishable. With MCA, target activations become more concentrated through timestep-aware conditional modulation. MSA further enhances target-related responses through frequency-guided memory selection and alignment, reducing residual background interference. These results demonstrate that the proposed modules provide discriminative conditional guidance for progressive denoising, enabling iterative refinement of uncertain target predictions and more reliable target localization.

2) Impact of backbone.: Table III reports the ablation results of different SAM2 Hiera backbone scales and adapters. Compared with the model without adapters, introducing adapters improves IoU by 3.23% and 1.96%, and P_d by 2.68% and 2.93% on SIRST-UAVB and SIRST4, respectively, demonstrating the effectiveness of infrared-domain adaptation. Increasing the backbone scale further enhances detection performance. Specifically, SAM2-small improves IoU to 72.34% and 80.42%, while SAM2-base achieves the highest IoU. These improvements indicate that larger SAM2 backbones provide stronger semantic representations and structural priors, offering more discriminative conditional guidance for recovering target details during progressive denoising. However, larger backbones introduce substantially more parameters. Therefore, SAM2-tiny is adopted as the default backbone to achieve a favorable balance between detection accuracy and computational efficiency.



(a) SIRST4 (b) SIRST-UAVB

Fig. 10: ROC curves of different methods on SIRST4 and SIRST-UAVB.

TABLE II: Ablation study of the proposed PDDiff on the SIRST-UAVB and SIRST4 datasets.

Settings	Denoising Network	DMSP	MCA	MSA	SIRST-UAVB				SIRST4			
					IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓
Baseline	–	–	–	–	62.35	88.28	78.05	4.56	73.41	92.26	81.72	19.91
w Denoising	✓	–	–	–	67.19	93.50	79.76	5.13	76.53	94.35	84.60	11.49
w DMSP	✓	✓	–	–	69.74	95.06	80.88	5.65	77.19	94.77	86.17	9.79
w MCA	✓	✓	✓	–	71.23	96.39	82.91	4.48	78.81	97.06	87.75	8.27
w MSA	✓	✓	✓	✓	**71.96**	**96.80**	**83.70**	**2.53**	**79.45**	**97.14**	**88.55**	**6.34**

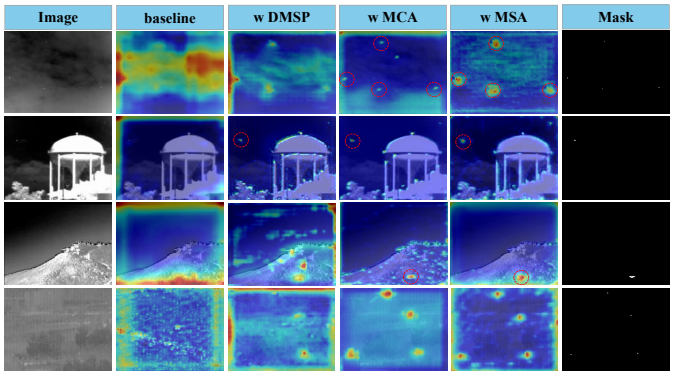


Fig. 11: Impact of different variants of PDDiff on modeling feature representation from SIRST4.

TABLE III: Ablation study of different Hiera backbone scales and adapters.

Settings	SIRST-UAVB				SIRST4				Params (M)
	IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓	
w/o Adapter	68.73	94.12	79.66	8.93	77.49	94.21	86.20	9.73	**42.96**
SAM2-tiny	71.96	96.80	83.70	2.53	79.45	97.14	88.55	6.34	43.71
SAM2-small	72.34	**96.89**	83.86	**2.29**	80.42	97.93	88.81	**6.12**	52.16
SAM2-base	**72.95**	96.74	**84.25**	5.11	**83.07**	**98.12**	**89.02**	9.55	238.71

3) Impact of training and sampling strategy.: Table IV reports the ablation results of the proposed training and sampling strategies. Introducing SMSP improves IoU by 1.74% and 2.33%, and P_d by 2.95% and 2.28% on SIRST-UAVB and SIRST4, respectively. By introducing structural perturbations into sparse target masks, SMSP encourages the denoising network to correct inaccurate target structures, thereby enhancing structure-aware denoising. As illustrated in Fig. 12, SMSP simulates structural deviations in sparse target masks through contour modification, erosion, dilation, and spatial shifting. These perturbations expose the model to diverse target distortions, improving its ability to recover accurate target structures during progressive denoising. Meanwhile, TPA improves IoU by 0.91% and 1.68%, while reducing F_a by 4.98 and 1.37 on the two datasets, respectively. This demonstrates that aggregating intermediate predictions effectively suppresses unstable responses and reduces false alarms. Combining SMSP and

TABLE IV: Ablation study of different training and sampling strategies.

Settings	SIRST-UAVB				SIRST4			
	IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓
Baseline	69.20	93.82	79.95	8.74	76.56	94.84	86.19	7.91
w SMSP	70.94	96.77	82.60	5.59	78.89	97.12	87.82	7.33
w TPA	70.11	95.62	82.14	3.76	78.24	95.79	88.16	6.54
w SMSP + TPA	**71.96**	**96.80**	**83.70**	**2.53**	**79.45**	**97.14**	**88.55**	**6.34**

TABLE V: Ablation study of different sampling steps and inference speed.

Time steps	SIRST-UAVB				SIRST4				FPS↑
	IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓	
$T = 5$	69.27	94.42	81.55	14.78	78.06	95.71	86.13	17.93	35.19
$T = 10$	70.55	96.14	83.38	7.94	78.86	96.78	87.31	10.19	28.33
$T = 12*$	71.96	96.80	83.70	2.53	79.45	97.14	88.55	6.34	25.74
$T = 15$	**72.25**	**96.91**	**83.74**	**2.51**	**79.72**	**97.45**	**88.61**	**6.10**	19.72

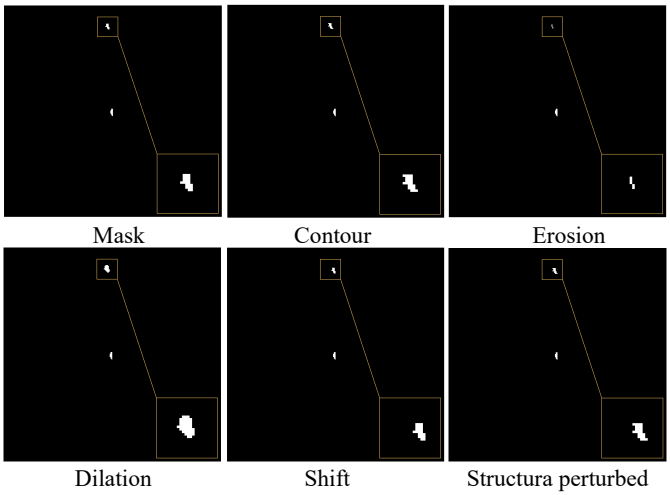


Fig. 12: Illustration of the SMSP strategy. Different structural perturbations, including contour modification, erosion, dilation, and spatial shifting, are applied to the original target mask to simulate structural distortions during diffusion training.

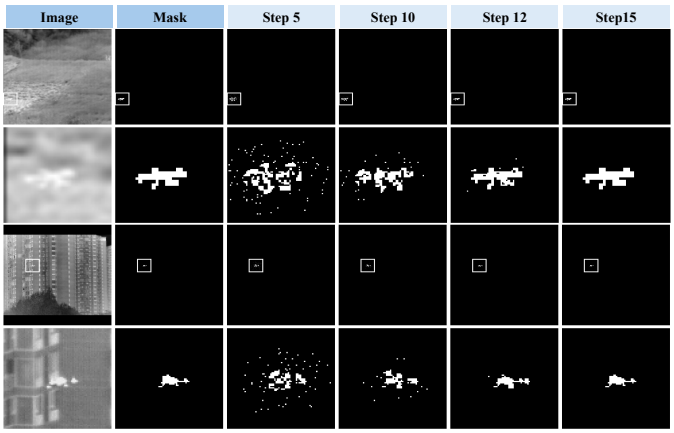


Fig. 13: Visualization of target mask refinement under different diffusion sampling steps. PDDiff progressively suppresses spurious responses and recovers accurate target structures as the number of sampling steps increases from 5 to 15.

TPA achieves the best overall performance on both datasets, demonstrating the complementary benefits of training and prediction aggregation.

4) Effectiveness of Different Time Steps: To investigate the impact of sampling steps on segmentation performance, we evaluate PDDiff with $T = 5$, 10, 12, and 15. As reported in Table V, increasing the sampling steps consistently improves detection accuracy on both datasets. Compared with $T = 5$, $T = 12$ improves IoU by 2.69% and 1.39%, while reducing F_a by 12.25 and 11.59 on SIRST-UAVB and SIRST4, respectively. These improvements demonstrate that iterative sampling facilitates progressive correction of uncertain mask predictions. However, further increasing T to 15 yields only marginal IoU gains of 0.29% and 0.27%, respectively, indicating diminishing benefits from additional denoising iterations. Meanwhile, as illustrated in Fig. 13, increasing the sampling steps gradually refines target structures and produces more accurate masks. Therefore, we adopt $T = 12$ as the default sampling setting to achieve a favorable balance between detection accuracy and inference efficiency.

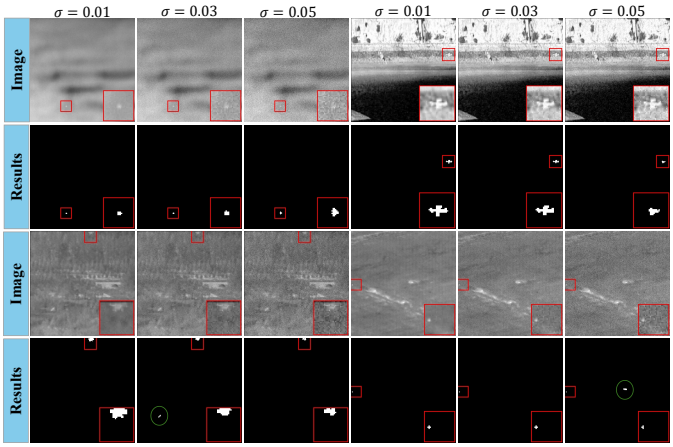


Fig. 14: Visual comparison of PDDiff under different Gaussian noise levels. Detection results are presented under increasing noise intensities ($\sigma = 0.01, 0.03$, and 0.05). Red boxes highlight target regions, while green circles indicate false alarms. The results illustrate the robustness of PDDiff in preserving weak target responses and suppressing background interference under varying noise conditions.

TABLE VI: Impact of different noise levels on SIRST4.

Methods	$\sigma = 0.01$			$\sigma = 0.03$			$\sigma = 0.05$		
	IoU↑	P_d ↑	F_a ↓	IoU↑	P_d ↑	F_a ↓	IoU↑	P_d ↑	F_a ↓
DNANet	74.82	94.78	20.16	70.60	90.24	28.41	52.85	84.50	47.04
MSHNet	73.33	94.19	22.12	71.19	90.52	26.35	56.06	85.48	44.76
SCTransNet	72.89	94.83	19.65	69.95	91.61	27.29	57.67	86.13	40.59
IRPNet	76.25	96.08	18.27	72.90	91.94	25.83	54.81	84.79	51.14
Ours	**77.83**	**96.63**	**14.75**	**75.54**	**93.19**	**23.52**	**61.26**	**87.41**	**37.91**

5) Impact of different noise level: To evaluate the robustness of PDDiff against imaging noise, we conduct experiments on SIRST4 under three Gaussian noise levels ($\sigma = 0.01$, 0.03, and 0.05), as reported in Table VI. As the noise level increases, all methods exhibit performance degradation due to increasingly severe noise interference. Nevertheless, PDDiff consistently achieves the best performance, outperforming the second-best methods by 1.58%, 1.64%, and 2.49% in IoU across the three noise levels, respectively. At $\sigma = 0.05$, PDDiff further improves IoU by 6.45% compared with IRPNet, demonstrating its robustness under stronger noise interference. As illustrated in Fig. 14, PDDiff can still accurately identify weak targets as the noise level increases, maintaining relatively stable target predictions. However, occasional false alarms emerge under severe noise interference, indicating that distinguishing weak targets from noise-induced responses remains challenging. These results suggest that iterative diffusion sampling facilitates progressive refinement of uncertain target predictions, reducing confusion caused by imaging noise.

6) TopK in MSA: We further investigate the influence of Top$K$ retrieval in MSA. With the memory bank sizes fixed to $[256, 128, 64, 32]$ across the four feature levels, we evaluate different retrieval settings, as reported in Table VII. Specifically, $[8, 8, 8, 4]$ achieves the best overall performance.

TABLE VII: Ablation study of different Top$K$ retrieval settings in MSA.

Settings	Top-K retrieval	SIRST-UAVB				SIRST4			
		IoU↑	P_d ↑	F_1 ↑	F_a ↓	IoU↑	P_d ↑	F_1 ↑	F_a ↓
(1)	[4, 4, 4, 2]	71.76	96.41	83.29	3.74	79.17	97.02	88.48	6.83
(2)	[8, 8, 8, 4]	**71.96**	**96.80**	**83.70**	**2.53**	**79.45**	**97.14**	**88.55**	**6.34**
(3)	[16, 16, 8, 4]	71.84	96.65	83.68	2.83	79.23	96.88	88.37	6.51

Reducing the retrieval number to $[4, 4, 4, 2]$ slightly degrades IoU by 0.28%, indicating that insufficient memory retrieval limits the exploitation of representative target patterns. In contrast, increasing it to $[16, 16, 8, 4]$ also reduces IoU by 0.22% in SIRST4 dataset, suggesting that retrieving excessive memory patches may introduce irrelevant background information. These results demonstrate that a moderate Top$K$ setting effectively balances relevant feature retrieval, providing more discriminative frequency-guided conditional representations for progressive denoising.

7) Limitations: Despite its promising performance, PDDiff still has several limitations. First, as a diffusion-based framework, PDDiff requires iterative sampling to progressively refine target predictions, introducing additional computational overhead compared with conventional single-step methods. As shown in Table V, increasing the sampling steps from $T = 12$ to $T = 15$ yields accuracy improvements while increasing inference cost. This indicates a trade-off between detection accuracy and computational efficiency. Future work will explore adaptive timestep scheduling and accelerated sampling strategies to reduce redundant denoising iterations.

Second, although DMSP and MSA enhance spatial-frequency representations, their effectiveness may be limited under extremely weak target responses and severe background interference. Specifically, the multiscale filtering in DMSP may suppress subtle target details, while the frequency memory selection in MSA may retrieve less discriminative features when target and background responses exhibit similar frequency characteristics. Future research will investigate target-preserving spatial filtering and more robust frequency-guided memory selection to improve detection reliability in challenging infrared scenes.

V. Conclusion

In this paper, we presented Progressive Denoising Diffusion (PDDiff), a conditional diffusion framework for infrared small target detection. PDDiff establishes a progressive generative denoising paradigm to address ambiguous target responses and background interference in conventional single-step prediction methods. In the conditional generation stage, DMSP enhances target-sensitive spatial representations through dynamic multiscale projection, while MSA selectively retrieves relevant memory features through frequency-domain similarity and phase alignment to provide discriminative conditional guidance. Furthermore, SMSP and TPA are introduced to improve sparse target structure learning during training and intermediate prediction aggregation during sampling, respectively. Extensive experiments on benchmark datasets demonstrate the effectiveness of PDDiff in terms of detection accuracy, noise robustness, and false-alarm suppression. Ablation studies further validate the contributions of the proposed components and strategies. These results indicate that progressive generative denoising provides an effective approach for reliable infrared small target detection. Future work will explore adaptive noise modeling and efficient sampling strategies to further improve detection robustness and computational efficiency.

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